Program 2: Working with Maven

Step 1: To create a Maven project run the following command in terminal:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

This command generates a basic Maven project structure with a sample App.java and AppTest.java.

```
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMo
de=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/MyMavenApp
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  20.035 s
[INFO] Finished at: 2026-03-17T22:29:18+05:30
[INFO] ------------------------------------------------------------------------
```

This “BUILD SUCCESS” will confirm that the project is created successfully

Step 2: Go to the project directory using command **cd MyMavenApp**

Step 3: Open and edit pom.xml file using command **gedit pom.xml**

- Go to https://mvnrepository.com
- Search for a dependency “junit”
- Select the latest version and copy the dependency XML snippet.
- Paste it under the <dependencies> section in pom.xml
- Updated pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MyMavenApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>
<name>MyMavenApp</name>
<url>http://maven.apache.org</url>
<!-- Java version configuration -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
```

```
<maven.compiler.target>11</maven.compiler.target>
</properties>
<!-- Dependencies -->
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!-- Build Plugins -->
<build>
<plugins>
<!-- Compiler Plugin -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>11</source>
<target>11</target>
</configuration>
</plugin>
<!-- Surefire Plugin for Testing -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<!-- JAR Plugin with Entry Point -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
```

```
</project>
```

edit App.java:

```
package com.example;
/**
 * Hello world!
 *
 */
public class App
{
   public static void main( String[] args )
   {
      System.out.println( "Hello World!" );
   }
}
```

Step 4: Build the project using command **mvn clean install**

```
shreya@ShreyaM:~/devops/MyMavenApp/src$ cd ..
shreya@ShreyaM:~/devops/MyMavenApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.024 s
[INFO] Finished at: 2026-03-17T19:33:48+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenApp$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.023 s - in com.example.AppTest
[INFO]
[INFO] Results:
```

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  7.901 s
[INFO] Finished at: 2026-03-17T19:34:06+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenApp ---
[INFO] Deleting /home/shreya/devops/MyMavenApp/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 2 source files to /home/shreya/devops/MyMavenApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/shreya/devops/MyMavenApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] -------------------------------------------------------
```

The “Build Success” message ensures that our Maven project is built successfully.

Step 5: To run the Maven application type he command:

java -jar target/MyMavenApp-1.0-SNAPSHOT.jar

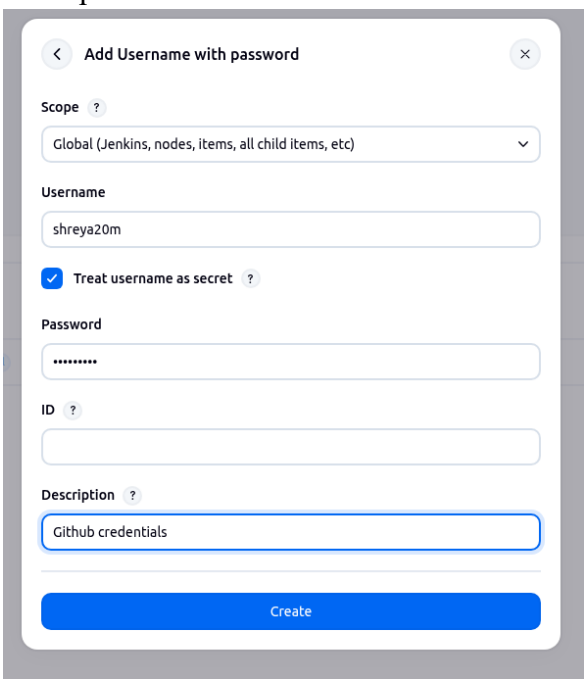
- After completing the execution, the output will look like:

```
[INFO] Compiling 1 source file to /home/shreya/devops/MyMavenApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenApp ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.049 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenApp ---
[INFO] Building jar: /home/shreya/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenApp ---
[INFO] Installing /home/shreya/devops/MyMavenApp/target/MyMavenApp-1.0-SNAPSHOT.jar to /home/shreya/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/shreya/devops/MyMavenApp/pom.xml to /home/shreya/.m2/repository/com/example/MyMavenApp/1.0-SNAPSHOT/MyMavenApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  14.777 s
[INFO] Finished at: 2026-03-17T19:34:37+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops/MyMavenApp$ java -jar target/MyMavenApp-1.0-SNAPSHOT.jar
Hello World!
shreya@ShreyaM:~/devops/MyMavenApp$
```

This will show output as the code present in App.java file we can edit the text to get any other output.

Continuous Integration (CI) Pipeline with Jenkins & Maven/Gradle

Step 1: Login to Jenkins and go to the Credentials page and click on add credentials
Step 2: Add new credentials in Jenkins for your CI pipeline, type such as "Username with password" for Git , enter details like username, password, set the scope to Global, and provide a description GitHub Credential.



Finally, click "Create" to securely store and use the credential in Jenkins pipelines. **Step 3:** now open the terminal and type the command **cd MyMavenApp** to enter the folder and create a **Jenkinsfile** .Paste the code in the Jenkinsfile

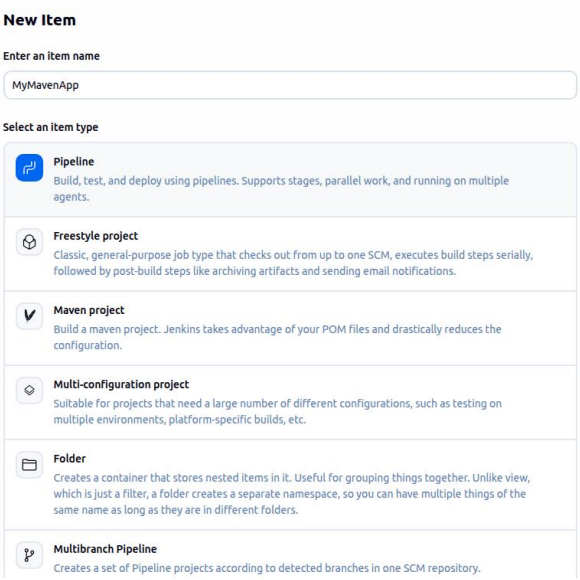
```
pipeline {
    agent any  // Use any available agent
    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/shreya20m /MyMavenApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
```

```
        }
    }
    stage('Test') {
        steps {
            sh 'mvn test'  // Run unit tests
        }
    }
    stage('Run Application') {
        steps {
            // Start the JAR application
            sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
        }
    }
}
post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }}}
```

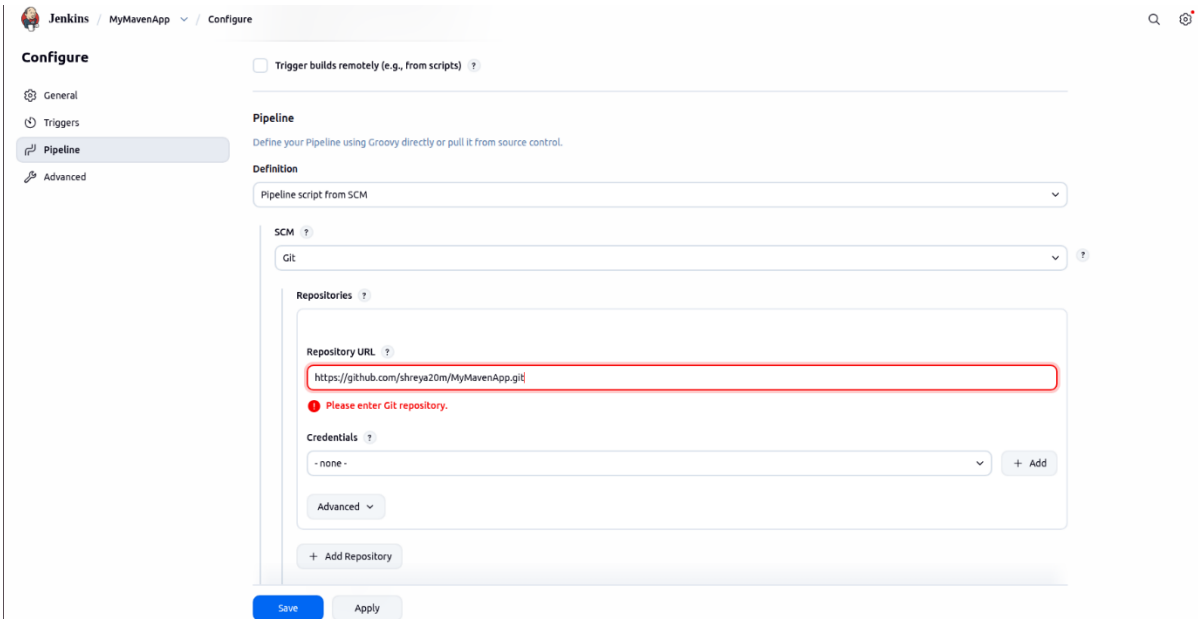
Then commit and push it to GitHub

```
shreya@ShreyaM:~/devops/MyMavenApp$ git remote set-url origin https://github.com
/shreya20m/MyMavenApp.git
shreya@ShreyaM:~/devops/MyMavenApp$ git push origin master
Username for 'https://github.com': shreya20m
Password for 'https://shreya20m@github.com':
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 1.39 KiB | 355.00 KiB/s, done.
Total 8 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/shreya20m/MyMavenApp.git
   3532cc9..c8f81f4  master -> master
shreya@ShreyaM:~/devops/MyMavenApp$
```

Step 4:Go to Jenkins Dashboard and click on new item , enter the name as MyMavenApp and choose pipeline



Step 5: In MyMavenApp, go to the Pipeline section and under Definition, select Pipeline script from SCM. Choose git ,type git repository url choose git credential which we created. Then click Apply and Save to finalize the configuration.



Step 6: in Jenkins and click Build Now. if errors appear under Build, Test, or Run Application, open your GitHub account, go to the MyMavenApp repository, and edit the pom.xml file to update the Java version change from 1.7 to 11

MyMavenApp - Stage View

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Average stage times:	3s	1s	2s	28s	439ms	174ms	263ms
#1 Apr 09 11:45 No Changes	3s	1s	2s	28s failed	439ms failed	174ms failed	263ms

After updating build again and check.

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Average stage times: (full run time: ~5min 3s)	13s	3s	5s	1min 9s	3s	12s	605ms
#2 Apr 14 13:30 1 commit	13s	4s	6s	2min 18s	5s	4s	719ms